

Question No : 1 of 26

Marks: 1 (Budgeted Time 1 Min)

ATM was designed for _____ and use a _____

Answer (Please select your correct option)

- ☐ LANs, Connectionless service paradigm
- ☐ WANs, Connectionless service paradigm
- ☐ Both LANs and WANs, Connectionless service paradigm
- ☐ Both LANs and WANs, Connection-oriented service paradigm

cornet Made By: Waqar Siddhu

Question No : 2 of 26

Marks: 1 (Budgeted Time 1 Min)

In broadcast topology there are two categories_____.

Answer (Please select your correct option)

☐ Satellite/Radio and Ring

☐ Satellite/Radio and Star

☐ Ring and star

☐ Satellite/Radio

correct

Made By: Waqar Siddhu

Question No : 3 of 26

Marks: 1 (Budgeted Time 1 Min)

----- have advantages arisen from the size and ease of computation.

Answer (Please select your correct option)

☐ CRC

☐ Parity

☐ Checksums

correct

☐ VRC

Made By: Waqar Siddhu

Question No : 4 of 26

Marks: 1 (Budgeted Time 1 Min)

When an application----- data, it makes a copy of the data available to all other computers on the network.

Answer (Please select your correct option)

☐ Broadcasting

☐ Multicasting

correct

☐ Unicasting

☐ Anycasting

Made By: Waqar Siddhu

Question No : 5 of 26

Marks: 1 (Budgeted Time 1 Min)

Basic LAN technologies such as Ethernet, Token Ring, and FDDI use a _____.

Answer (Please select your correct option)

- ☐ Connectionless service paradigm
- ☐ Connection-oriented service paradigm
- ☐ Both Connectionless and Connection-oriented service paradigm
- ☐ Connection-oriented service paradigm without acknowledgment

correct

Made By: Waqar Siddhu

Question No : 6 of 26

Marks: 1 (Budgeted Time 1 Min)

If the sending computer crashes before transmitting a complete frame, the _____ will not arrive and the receiver will know that the frame is _____.

Answer (Please select your correct option)

☐ soh, complete

☐ soh, incomplete

☐ eot, incomplete

correct

☐ eot, complete

Made By: Waqar Siddhu

Question No : 7 of 26

Marks: 1 (Budgeted Time 1 Min)

----- was especially most concerned about the lack of highly powered computers.

Answer (Please select your correct option)

☐ ARPA

correct

☐ IEEE

☐ EIA

☐ Government

Made By: Waqar Siddhu

Question No : 8 of 26

Marks: 1 (Budgeted Time 1 Min)

"Frame type" field of the header consists of _____ bits.

Answer (Please select your correct option)

☐ 8

☐ 10

☐ 16

☐ 32

correct

Made By: Waqar Siddhu

Question No : 9 of 26

Marks: 1 (Budgeted Time 1 Min)

Virtual Circuit connection identifier occupies _____ bits.

Answer (Please select your correct option)

☐ 16

correct

☐ 24

☐ 32

☐ 8

Made By: Waqar Siddhu

Question No : 10 of 26

Marks: 1 (Budgeted Time 1 Min)

_____ senses a signal on a cable and then transmits an amplified copy of the signal.

Answer (Please select your correct option)

- ☐ Bridge
- ☐ Hub
- ☐ Transceiver
- ☐ Repeater

correct

Made By: Waqar Siddhu

Question No : 11 of 26

Marks: 1 (Budgeted Time 1 Min)

One repeater _____, two repeaters _____ the maximum cable length limitation.

Answer (Please select your correct option)

☐ Square root, Cube root

☐ Double, Triple

☐ Triple, Quad

☐ Double, Quad

correct

Made By: Waqar Siddhu

Question No : 12 of 26

Marks: 1 (Budgeted Time 1 Min)

A Bridge forwards or filters a frame by comparing the information in its address table to the frame's _____.

Answer (Please select your correct option)

☐ Layer 2 source address

☐ Layer 3 source address

☒ Layer 2 destination address

☐ Layer 3 destination address

correct

Made By: Waqar Siddhu

Question No : 13 of 26

Marks: 1 (Budgeted Time 1 Min)

A network that permits many sources to take turns accessing a shared communication resource is providing a form of _____.

Answer (Please select your correct option)

- ☐ Phase Division Multiplexing
- ☐ Time Division Multiplexing
- ☐ Frequency Division Multiplexing
- ☐ Wave Division Multiplexing

Made By: Waqar Siddhu

Question No : 14 of 26

Marks: 1 (Budgeted Time 1 Min)

No error detection scheme is perfect because transmission errors can affect _____.

Answer (Please select your correct option)

☐ Data

☐ Additional Information

☐ Additional information & data

☐ Transmission medium

correct

Made By: Waqar Siddhu

Question No : 15 of 26

Marks: 1 (Budgeted Time 1 Min)

LAN that uses ATM technology has a _____.

Answer (Please select your correct option)

☐ Star topology

correct

☐ Bus topology

☐ Ring topology

☐ Mesh topology

Made By: Waqar Siddhu

Question No : 16 of 26

Marks: 1 (Budgeted Time 1 Min)

FDDI can transmits data at a rate of _____.

Answer (Please select your correct option)

☐ 100 million bits per second

correct

☐ 10 million bits per second

☐ 1000 million bits per second

☐ 3 million bits per second

Made By: Waqar Siddhu

Question No : 17 of 26

Marks: 1 (Budgeted Time 1 Min)

The next hop to which a packet is sent depends only on_____.

Answer (Please select your correct option)

☐ Packet's destination

correct

☐ Packet's original source

☐ Path the packet has taken

☐ Routed Protocol

Made By: Waqar Siddhu

Question No : 18 of 26

Marks: 1 (Budgeted Time 1 Min)

Most WAN systems include a mechanism that can be used to eliminate the common case of duplication routing is called _____.

Answer (Please select your correct option)

☐ Dynamic route

☐ Hierarchal address

☐ Default route

☐ Shortest path

correct

Made By: Waqar Siddhu

Question No : 19 of 26

Marks: 1 (Budgeted Time 1 Min)

_____ in which delay among packets can vary.

Answer (Please select your correct option)

☐ Virtual Private Network

☐ Isochronous Networks

☐ Asynchronous Networks

☐ Synchronous Networks

Made By: Waqar Siddhu

Question No : 20 of 26

Marks: 1 (Budgeted Time 1 Min)

A typical port on an ATM switch operates at _____ or higher.

Answer (Please select your correct option)

☐ OC-2 speed (155Mbps)

☐ OC-3 speed (100Mbps)

☐ OC-3 speed (155Mbps)

☐ OC-3 speed (155Gbps)

correct

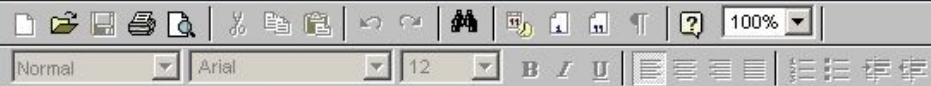
Made By: Waqar Siddhu

Question No : 21 of 26

Marks: 2 (Budgeted Time 4 Min)

Suppose XYZ organization is using hierarchal addressing scheme in routing tables. Write any two advantages of using this scheme of addressing.

Answer ([Please click here to Add Answer](#))



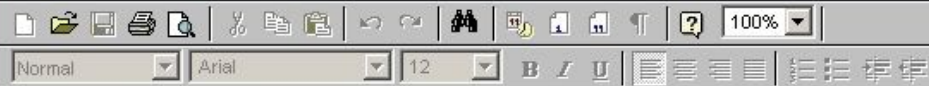
Made By: Waqar Siddhu

Question No : 22 of 26

Marks: 2 (Budgeted Time 4 Min)

Suppose ABC Corporation has a small number of packet switches and computer systems in its computer network. If entire network is stable with single network link than which routing method will be most appropriate for this scenario?

Answer ([Please click here to Add Answer](#))



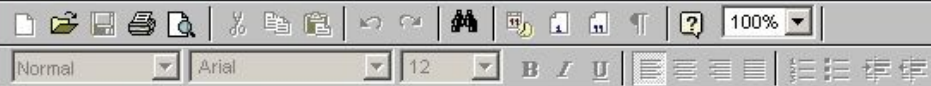
Made By: Waqar Siddhu

Question No : 23 of 26

Marks: 3 (Budgeted Time 6 Min)

Suppose in university campus, bridges are used in its local area network. Explain all steps, how bridge builds up addresses list table of all attached computers?

Answer ([Please click here to Add Answer](#))



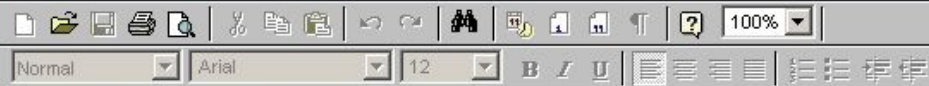
Made By: Waqar Siddhu

Question No : 24 of 26

Marks: 3 (Budgeted Time 6 Min)

Suppose you are a Network Engineer in an ISP which is providing different technological services like one of these may be cell based network. Here, as a Network Engineer you must know that which term is used to refer the “information” related to the network but not refers to the “information” that only uses the network”. Give the name of this term and the reserved values it uses.

Answer ([Please click here to Add Answer](#))



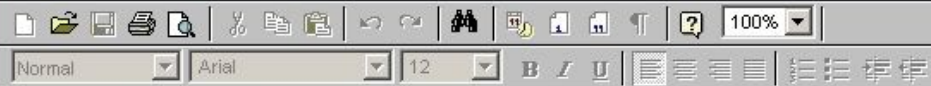
Made By: Waqar Siddhu

Question No : 25 of 26

Marks: 5 (Budgeted Time 10 Min)

A network sends and receives bits faster than computer's CPU. Explain how a computer attached to a network can handle this speed difference? Does the CPU have any role in handling it?

Answer ([Please click here to Add Answer](#))



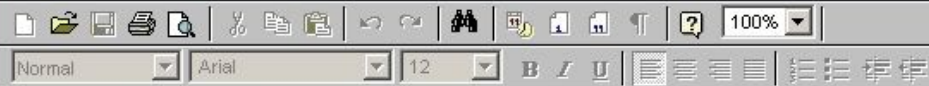
Made By: Waqar Siddhu

Question No : 26 of 26

Marks: 5 (Budgeted Time 10 Min)

Star textile has multiple LANs with packet switches. It is critical that if some link goes down/breakup the entire network traffic should arrive to respective destination smoothly. So, which method will meet this requirement? Give at least three reasons to support your answer.

Answer ([Please click here to Add Answer](#))



Made By: Waqar Siddhu

Question No : 1 of 26

Marks: 1 (Budgeted Time 1 Min)



In -----, network occupies the smaller area like a room a floor or a building

Answer (Please select your correct option)

☐ LAN

correct

☐ WAN

☐ MAN

☐ PAN

Made By: Waqar Siddhu

Question No : 2 of 26

Marks: 1 (Budgeted Time 1 Min)



A network that uses a ----- usually consist of a single long cable to which computers attach.

Answer (Please select your correct option)

☐ Bus Topology

correct

☐ Tree Topology

☐ Star Topology

☐ Ring Topology

Made By: Waqar Siddhu

Question No : 3 of 26

Marks: 1 (Budgeted Time 1 Min)



A network uses a ----- if all computers attach to a central point

Answer (Please select your correct option)

☐ Star Topology

correct

☐ Ring Topology

☐ Bus Topology

☐ Mesh Topology

Made By: Waqar Siddhu

Question No : 4 of 26

Marks: 1 (Budgeted Time 1 Min)



A----- provides mechanism that automatically assigns a physical address to a station when the station first boots.

Answer (Please select your correct option)

- ☐ Static addressing scheme
- ☐ Configurable addressing scheme
- ☒ Dynamic addressing scheme
- ☐ Dynamic Host Configuration

correct

Made By: Waqar Siddhu

Question No : 5 of 26

Marks: 1 (Budgeted Time 1 Min)

----- was especially most concerned about the lack of highly powered computers.

Answer (Please select your correct option)

☐ ARPA

correct

☐ IEEE

☐ EIA

☐ Government

Made By: Waqar Siddhu

Question No : 6 of 26

Marks: 1 (Budgeted Time 1 Min)



A bridge functions in the _____ layer(s).

Answer (Please select your correct option)

☐ Physical

correct

☐ Data link

☐ Network and Data link

☐ Physical and Data link

Made By: Waqar Siddhu

Question No : 7 of 26

Marks: 1 (Budgeted Time 1 Min)



Which of the following is not used to extend LAN ?

Answer (Please select your correct option)

- ☐ Bridge
- ☐ Repeater
- ☐ Fiber modem
- ☐ Router

correct

Made By: Waqar Siddhu

Question No : 8 of 26

Marks: 1 (Budgeted Time 1 Min)



The working of connection-oriented service for networking is similar to the _____.

Answer (Please select your correct option)

☐ Ethernet

☐ Telephone

correct

☐ IP

☐ UDP

Made By: Waqar Siddhu

Question No : 9 of 26

Marks: 1 (Budgeted Time 1 Min)



_____ senses a signal on a cable and then transmits an amplified copy of the signal.

Answer (Please select your correct option)

☐ Repeater

correct

☐ Bridge

☐ Hub

☐ Transceiver

Made By: Waqar Siddhu

Question No : 10 of 26

Marks: 1 (Budgeted Time 1 Min)



Unlike Frame Relay and ATM, SMDS (Switched multi-megabit Data service) offers_____.

Answer (Please select your correct option)

☐ Connectionless service paradigm

correct

☐ Connection oriented service paradigm

☐ Both Connectionless and Connection-oriented service paradigm

☐ Stateless Connection oriented service paradigm

Made By: Waqar Siddhu

Question No : 11 of 26

Marks: 1 (Budgeted Time 1 Min)



A Bridge forwards or filters a frame by comparing the information in its address table to the frame's _____.

Answer (Please select your correct option)

☐ Layer 2 source address

☐ Layer 3 source address

☒ Layer 2 destination address

☐ Layer 3 destination address

correct

Made By: Waqar Siddhu

Question No : 12 of 26

Marks: 1 (Budgeted Time 1 Min)



ATM is designed to work on_____.

Answer (Please select your correct option)

☐ Twisted Pair

☐ Coaxial

☐ Radio Frequency

☐ Fiber

correct

Made By: Waqar Siddhu

Question No : 13 of 26

Marks: 1 (Budgeted Time 1 Min)



_____ identifies intermediate computers along a path to remote destination

Answer (Please select your correct option)

- ☐ Ping
- ☐ Traceroute
- ☐ ICMP
- ☐ OSPF

correct

Made By: Waqar Siddhu

Question No : 14 of 26

Marks: 1 (Budgeted Time 1 Min)



_____ computes shortest paths in a graph by using weights on edges as a measure of distance.

Answer (Please select your correct option)

☐ Greedy Algorithm

☐ Distance vector Algorithm

☐ Dijkstra's algorithm

☐ Binary Search Algorithm

correct

Made By: Waqar Siddhu

Question No : 15 of 26

Marks: 1 (Budgeted Time 1 Min)



_____ has no way to determine the cause of the problem.

Answer (Please select your correct option)

☐ ICMP

☐ ARP

☐ Ping

☐ Traceroute

correct

Made By: Waqar Siddhu

Question No : 16 of 26

Marks: 1 (Budgeted Time 1 Min)



A network that permits many sources to take turns accessing a shared communication resource is providing a form of _____.

Answer (Please select your correct option)

- ☐ Phase Division Multiplexing
- ☐ Time Division Multiplexing
- ☐ Frequency Division Multiplexing
- ☐ Wave Division Multiplexing

Made By: Waqar Siddhu

Question No : 17 of 26

Marks: 1 (Budgeted Time 1 Min)



No error detection scheme is perfect because transmission errors can affect _____.

Answer (Please select your correct option)

☐ Additional Information

☐ Additional information & data

☐ Transmission medium

☐ Data

correct

Made By: Waqar Siddhu

Question No : 18 of 26

Marks: 1 (Budgeted Time 1 Min)



Computers attached to an ethernet use _____ in which a computer waits for the Ether to be idle before transmitting a frame.

Answer (Please select your correct option)

☐ CSMA/CD

correct

☐ CSMA/CA

☐ Token Passing

☐ Round Robin

Made By: Waqar Siddhu

Question No : 19 of 26

Marks: 1 (Budgeted Time 1 Min)



Most WAN systems include a mechanism that can be used to eliminate the common case of duplication routing is called_____.

Answer (Please select your correct option)

☐ Dynamic route

☐ Hierarchal address

☐ Default route

☐ Shortest path

correct

Made By: Waqar Siddhu

Question No : 20 of 26

Marks: 1 (Budgeted Time 1 Min)



A typical port on an ATM switch operates at _____ or higher.

Answer (Please select your correct option)

☐ OC-2 speed (155Mbps)

☐ OC-3 speed (100Mbps)

☐ OC-3 speed (155Mbps)

☐ OC-3 speed (155Gbps)

correct

Made By: Waqar Siddhu

Question No : 21 of 26

Marks: 2 (Budgeted Time 4 Min)



Network operator needs a temporary virtual circuit for his network. Which type of ATM virtual circuit is most appropriate in this situation?

Answer ([Please click here to Add Answer](#))



Made By: Waqar Siddhu

Question No : 22 of 26

Marks: 2 (Budgeted Time 4 Min)

Default routes are used in networks. You are required to write two benefits of using default route.

Answer ([Please click here to Add Answer](#))



Made By: Waqar Siddhu

Question No : 23 of 26

Marks: 3 (Budgeted Time 6 Min)



ATM provides the best support for multiple services. Due to which reasons ATM is not adopted as a single underline technology? Give any three reasons to justify your answer.

Answer ([Please click here to Add Answer](#))



Made By: Waqar Siddhu

Question No : 24 of 26

Marks: 3 (Budgeted Time 6 Min)



If a packet switch does not check header's source information then how it can be beneficial for routing? Write any three benefits.

Answer ([Please click here to Add Answer](#))



Made By: Waqar Siddhu

Question No : 25 of 26

Marks: 5 (Budgeted Time 10 Min)

Suppose, as a Network Administrator of XYZ organization you are given a task to develop a network in a limited area with low cost, then which wiring scheme will you use? Justify your answer with four solid reasons.

Answer ([Please click here to Add Answer](#))



Made By: Waqar Siddhu

Question No : 26 of 26

Marks: 5 (Budgeted Time 10 Min)

Suppose there are two networks A and B both using virtual private network. If a user of some another network other than the network B wants to send some packets to a device in network A, then what will be the behavior of network A? Give three reasons behind its behavior.

Answer ([Please click here to Add Answer](#))



Made By: Waqar Siddhu

In _____, network occupies the smaller area like a room, a floor or a building.

Answer (Please select your correct option)

☐ LAN

correct

☐ WAN

☐ MAN

☐ Wireless WAN

Made By: Waqar Siddhu

In which type of network topology, computers are connected to a central device?

Answer (Please select your correct option)

☐ Ring

correct

☐ Mesh

☐ Bus

☐ Star

Made By: Waqar Siddhu

The term _____ refers to the general concept of a small block of data in a computer network.

Answer (Please select your correct option)

☐ Packet

correct

☐ Byte

☐ Cell

☐ Information

Made By: Waqar Siddhu

_____ is the scheme, which is designed to help in detecting transmission errors by sending one extra bit of information with each character.

Answer (Please select your correct option)

☐ Parity checking

correct

☐ Checksums

☐ CRC

☐ LRC

Made By: Waqar Siddhu

_____ have advantages arisen from the size and ease of computation.

Answer (Please select your correct option)

☐ CRC

☐ Parity

☒ Checksums

☐ VRC

correct

Made By: Waqar Siddhu

The Fast Ethernet hardware operates at a rate of _____.

Answer (Please select your correct option)

☐ 10 Mbps

☒ 100 Mbps

☐ 1000 Mbps

☐ 1 Gbps

correct

Made By: Waqar Siddhu

The Gigabit Ethernet hardware operates at a rate of _____.

Answer (Please select your correct option)

☐ 10 Mbps

☐ 100 Mbps

☒ 1000 Mbps

☐ 10 Gbps

correct

Made By: Waqar Siddhu

Wireless LAN uses _____ as a media access method.

Answer (Please select your correct option)

☐ CSMA/CD

☐ CSMA/CA

☐ Token Passing

☐ CSMA

correct

Made By: Waqar Siddhu

The _____ relies on the hardware manufacturer to assign a unique physical address to each network interface.

Answer (Please select your correct option)

☐ Static addressing scheme

correct

☐ Configurable addressing scheme

☐ Dynamic addressing scheme

☐ Dynamic Configurable addressing scheme

Made By: Waqar Siddhu

IEEE LLC/SNAP header is consist of _____ which is used to specify the type of data.

Answer (Please select your correct option)

☐ 8 octets

correct

☐ 10 octets

☐ 12 bytes

☐ 16 bits

Made By: Waqar Siddhu

Formally named _____ is informally known as the twisted pair Ethernet or TP Ethernet.

Answer (Please select your correct option)

☐ 10 Base 2

☐ 10 Base 5

☐ 10 Base T

☐ 100 Base T

correct

Made By: Waqar Siddhu

_____ senses a signal on a cable and then transmits an amplified copy of the signal.

Answer (Please select your correct option)

- ☐ Bridge
- ☐ Hub
- ☐ Transceiver
- ☐ Repeater

correct

Made By: Waqar Siddhu

Generally a Bridge is used to perform _____.

Answer (Please select your correct option)

☐ Extend a LAN, Filter and Forward a packet

☐ Routing

☐ Filter and Forward a frame

☐ Filter and Forward a packet

correct

Made By: Waqar Siddhu

Which of the following is not used to extend LAN ?

Answer (Please select your correct option)

- ☐ Bridge
- ☐ Repeater
- ☐ Fiber modem
- ☐ HUB

correct

Made By: Waqar Siddhu

If a bridge is unable to find _____ in its table then it just forwards frames to the other segment.

Answer (Please select your correct option)

☐ IP address

☐ Destination address

☐ Source address

☐ Port address

Made By: Waqar Siddhu

Most WAN systems include a mechanism that can be used to eliminate the common case of duplication routing is called_____.

Answer (Please select your correct option)

☐ Dynamic route

☐ Hierarchal address

☐ Default route

☐ Shortest path

correct

Made By: Waqar Siddhu

The working of connection-oriented service for networking is similar to the _____.

Answer (Please select your correct option)

☐ Ethernet

☒ Telephone

☐ IP

☐ UDP

correct

Made By: Waqar Siddhu

ATM assigns each VC a _____ identifier which is divided into two parts to produce a hierarchy.

Answer (Please select your correct option)

☐ 23-bit

☐ 24-bit

correct

☐ 21-bit

☐ 22-bit

Made By: Waqar Siddhu

_____ is used for delay sensitive applications requiring high data rate.

Answer (Please select your correct option)

☐ Constant Bit Rate (CBR) service

correct

☐ Variable Bit Rate (VBR) service

☐ Available Bit Rate (ABR) service

☐ Uniform Bit Rate (UBR) service

Made By: Waqar Siddhu

Header of cell takes _____ of network capacity.

Answer (Please select your correct option)

☐ 5%

☐ 10%

☐ 12%

☐ 15%

correct

Made By: Waqar Siddhu

Network operator needs a temporary virtual circuit for his network. Which type of ATM virtual circuit is most appropriate in this situation?

Answer ([Please click here to Add Answer](#))



Describe two major components of thick Ethernet.

Answer ([Please click here to Add Answer](#))



If a packet switch does not check header's source information then how it can be beneficial for routing? Write any three benefits.

Answer ([Please click here to Add Answer](#))



Is it necessary to use terminator at both ends of thin Ethernet cable? Give your answer with at least two reasons.

Answer ([Please click here to Add Answer](#))



Suppose, as a Network Administrator of XYZ organization you are given a task to develop a network in a limited area with low cost, then which wiring scheme will you use? Justify your answer with four solid reasons.

Answer ([Please click here to Add Answer](#))



Star textile has multiple LANs with packet switches. It is required that if any link goes down/breakup, the entire network traffic should arrive to respective destination smoothly. Which routing method will meet this requirement? Give at least three reasons to support your answer.

Answer ([Please click here to Add Answer](#))





In broadcast topology there are two categories_____.

Answer (Please select your correct option)



Satellite/Radio and Ring

correct



Satellite/Radio and Star



Ring and star



Satellite/Radio

Made By: Waqar Siddhu



In -----, network occupies the smaller area like a room a floor or a building

Answer (Please select your correct option)



LAN

correct



WAN



MAN



PAN

Made By: Waqar Siddhu



The Ethernet standard specifies that frames are sent using the -----

Answer (Please select your correct option)

☐ Manchester Encoding

correct

☐ Nonreturn to zero(NRZ)

☐ Return to zero(RZ)

☐ Differential Manchester

Made By: Waqar Siddhu



A ----- relies on the hardware manufacturer to assign a unique physical address to each network interface.

Answer (Please select your correct option)

☐ Static addressing scheme

correct

☐ Configurable addressing scheme

☐ Dynamic addressing scheme

☐ Burned MAC

Made By: Waqar Siddhu



When an application----- data, it makes a copy of the data available to all other computers on the network.

Answer (Please select your correct option)

☐ Broadcasting

☐ Multicasting

☐ Unicasting

☐ Anycasting

correct



Formally named _____ informally known as the thick wire Ethernet or Thick net.

Answer (Please select your correct option)

☐ 10 Base 2

☐ 10 Base 5

☐ 10 Base T

☐ 100 Base T

correct



Basic LAN technologies such as Ethernet, Token Ring, and FDDI use a _____.

Answer (Please select your correct option)

- ☐ Connectionless service paradigm
- ☐ Connection-oriented service paradigm
- ☐ Both Connectionless and Connection-oriented service paradigm
- ☐ Connection-oriented service paradigm without acknowledgment

correct

Made By: Waqar Siddhu



In which type of topology computers are connected to a central device?

Answer (Please select your correct option)

☐ Ring

correct

☐ Mesh

☐ Bus

☐ Star

Made By: Waqar Siddhu



----- was especially most concerned about the lack of highly powered computers.

Answer (Please select your correct option)

☐ ARPA

correct

☐ IEEE

☐ EIA

☐ Goverment

Made By: Waqar Siddhu



"Frame type" field of the header consists of _____ bits.

Answer (Please select your correct option)

8



10



16



32



correct

Made By: Waqar Siddhu



A bridge uses the _____ to determine which computers connect to which segment and uses the _____ to determine whether to forward a copy of frame.

Answer (Please select your correct option)



Source address, Destination address

correct



Destination address, Source address



Destination address, MAC Address



Destination address, Destination address

Made By: Waqar Siddhu



Generally a Bridge is used to perform _____.

Answer (Please select your correct option)



Extend a LAN, Filter and Forward a packet



Routing



Filter and Forward a frame



Filter and Forward a packet

correct

Made By: Waqar Siddhu



If bridge cannot find _____ in its tables then forwards frames to the other segment.

Answer (Please select your correct option)

☐

IP address

☐

Destination address

☐

Source address

☐

Port address

correct



_____ technology can support both LAN and WAN connections.

Answer (Please select your correct option)



Frame relay



Ethernet



SMDS



ATM

correct

Made By: Waqar Siddhu



Virtual Circuit connection identifier occupies _____ bits.

Answer (Please select your correct option)

☐ 16

correct

☐ 24

☐ 32

☐ 8

Made By: Waqar Siddhu



_____ is used for ordinary data.

Answer (Please select your correct option)



Available Bit Rate (ABR) service

correct



Uniform Bit Rate (UBR) service



Constant Bit Rate (CBR) service



Variable Bit Rate (VBR) service

Made By: Waqar Siddhu



Unlike Frame Relay and ATM, SMDS (Switched multi-megabit Data service) offers _____.

Answer (Please select your correct option)

- ☐ Connectionless service paradigm
- ☐ Connection oriented service paradigm
- ☐ Both Connectionless and Connection-oriented service paradigm
- ☐ Stateless Connection oriented service paradigm



The term _____ refers to the general concept of a small block of data in a network.

Answer (Please select your correct option)

☐ Packet

correct

☐ Byte

☐ Cell

☐ Information

Made By: Waqar Siddhu



LAN that uses ATM technology has a _____.

Answer (Please select your correct option)

☐ Star topology

correct

☐ Bus topology

☐ Ring topology

☐ Mesh topology



The next hop to which a packet is sent depends only on_____.

Answer (Please select your correct option)

☐ Packet's destination

correct

☐ Packet's original source

☐ Path the packet has taken

☐ Routed Protocol



Network operator needs a temporary virtual circuit for his network. Which type of ATM virtual circuit is most appropriate in this situation?

Answer ([Please click here to Add Answer](#))





How many methods are available to span a bridge network over long distances? Just write their names.

Answer ([Please click here to Add Answer](#))





Suppose in university campus, bridges are used in its local area network. Explain all steps, how bridge builds up addresses list table of all attached computers?

Answer ([Please click here to Add Answer](#))





Suppose XYZ Company has packet switch network of 50 computers. After some time of communication, network administrator observes that the packets in the packet switches are facing some delay just before these are transmitted over the link. Identify this delay and write down the mathematical equation of the relationship between delay and throughput.

Answer ([Please click here to Add Answer](#))





While designing the ATM technology why the designers of the ATM have chosen cells over packets which are being used widely? Give five reasons to support your answer.

Answer ([Please click here to Add Answer](#))





Suppose you are assigned to connect two site offices which are situated far away from each other. In your opinion, which one will be the appropriate solution for this, LAN extension or WAN? Justify your answer with four reasons.

Answer ([Please click here to Add Answer](#))





CS610- Computer Network

Solved Subjective
From Midterm Papers

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CS610- Computer Network Midterm Examination - Fall 2011

1. Where are destination and source address located in frame? (2 marks)

Answer:- (Page 56)

The packet header includes destination and source addresses.

2. If 100mbps Ethernet NIC is connected to a hub which supports 10mbps, what will be the data rate in this scenario? (2 marks)

Answer:-

This has two consequences. First, the hub does not necessarily create a broadcast medium. Second, there are no collisions between traffic on 10 Mbps ports and 100 Mbps ports.

3. In connection oriented network, which type of connection identifier is used? Support your answer with example. (2 marks)

Answer:- (Page 66)

ATM uses the concept of connection-oriented networking.

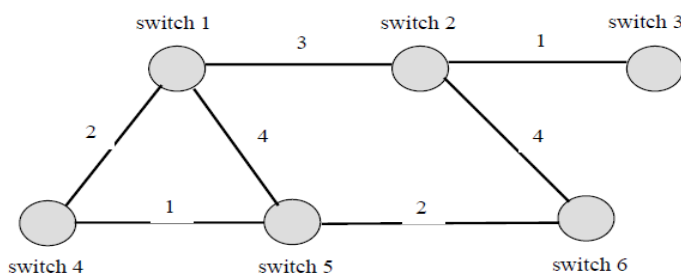
4. Imagine a network of 6 devices A, B, C, D, E and F. All devices are on same packet switch. If computer A wants to send packet to computer E, How would packet switch forward this packet? (3 marks)

Answer:-

path between switch A and F and the total distance traveled along this path

switch A – switch D – switch E

distance = 4



5. Network administrator needs audio and video service for his network. He has to send or receive data at same data rate. Which type of ATM quality service specification would be applied there?

Give example? (3 marks)

Answer:-

To facilitate the transport of audio and video data across emerging Asynchronous Transfer Mode (ATM) networks, a simple, low cost, audio/video ATM appliance, the AVATAR, has been developed. This appliance is capable of handling uncompressed bidirectional audio and NTSC video connections.

6. We have two satellite locations, at first location one bridge is configured and at second location 2 bridges are configured. Which location performed will be faster? Give reason. (3 marks)

Answer:- [Click here for detail](#)

Satellite locations with two bridges configured will performed faster. The brigade also had to provide power to virtually all of its East Timor assets. Setting up the satellite system, for example, required supplying power to both the communications station and the users' computers. A bridge that has multiple ports is known as a networking switch. Both bridges and switches are capable of directing traffic to specific network addresses instead of broadcasting the data to all devices on a network segment. This functionality makes the bridge or switches a more advanced networking device over a hub or repeater.

7. There are two sites office A & B, both are using VPN. If a user of another network wants to send some packet to A, then what will be the behavior of network A? How data is protected in VPN environment? (05 marks)

Answer:- [Click here for detail](#)

In theory when a machine on private network A wants to send a packet to a machine on private network B the packet is accepted by the VPN server. It is then encrypted and encapsulated within a standard IP packet that has a destination address corresponding to the VPN server at network B.

Notice that both VPN servers have to have to be “on the Internet” in the sense of having publicly accessible IP addresses. When the packet arrives at network B's VPN server it extracts the encapsulated private packet, decrypts it and sends it on its way within the local network. The machines within network A and B don't have to have publicly accessible address - that is they don't have to be on the public Internet. This way of using the public Internet to connect two machines that are not on the public internet is another advantage of the VPN approach.

As far as users of network A and B are concerned the VPN is as good as a dedicated physical connection. The idea of using public data packets to transfer embedded private data packets is often called “tunnelling” because it seems to create a data tunnel between the two networks.

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1. We have two satellite locations. At first location one bridge is configured and at 2nd location two bridges are configured. Which location's performance will be faster. Give reasons to support your answer. (5 marks)

Answer:- rep

2. Star Textile has huge network of systems and switches. A task is given to IT Officer that if some link goes the entire network works smoothly. Which routing method will be used? 5 marks

Answer:- [Click here for detail](#)

Distributed routing method is better for this purpose because Distributed routing relies on each node to compute its own routing table and build the required connections with its neighbors. Ideally, the network operation, status, and architecture of each node is transparent. Distributed routing is more flexible than centralized routing because each node handles its own routing. The result is often improved system performance.

3. Six devices are connected in network A,B,C,D,E and F. All devices are on same packet switch. Computer A wants to send packet to E. How will packet switch forwards the packet? 3 marks

Answer:- rep

4. Network administrator wants to build a tunnel between sites offices. How will he build using private virtual network? 3 marks

Answer:- [Click here for detail](#)

To establish a private network that can send data securely between these two locations or networks through a "tunnel." A VPN tunnel connects the two PCs or networks and allows data to be transmitted over the Internet as if it were still within those networks.

5. Who examines performance of network? 2 marks

Answer:- [Click here for detail](#)

Performance is probably the most closely monitored operational feature by users in the network environment

6. How can we compute shortest path in WAN? 2 marks

Answer:- (Page 62)

We use Dijkstra's algorithm to compute shortest path from each node to every other node.

7. 100Mbps Ethernet NIC is connected to control Hub which supports 10Mbps. What will be Data Rate in this scenario? 2 marks

CS610- Computer Network Midterm Examination – Fall 2011

There are two sites office A & B, both are using VPN. If a user of another network wants to send some packet to A, then what will be the behavior of network A?

Answer:- rep

How data is protected in VPN environment? (05 marks)

Answer:- rep

What type of switching technique is used in WAN?

Answer:- (Page 55)

WANs can use circuit-switching or packet-switching techniques. To span long distances or many computers, networks must replace shared medium with packet switches. Each switch moves an entire packet from one connection to another. That's why they are called packet switches.

write the names of identifiers used in ATM.(2)

What type of topology used in Ethernet? (2).

Answer:- [Click here for detail](#)

Traditional Ethernet employs a bus topology, meaning that all devices or hosts on the network use the same shared communication line.

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Difference between LAN and WAN 2 marks

Answer:- (Page 4)

In LAN network occupies the smaller area like a room a floor or a building.

In WAN, network occupies larger areas like cities & countries. Internet is a Wide Area Network.

Why fiber need fir ATM 3 marks

define bridge 3 marks

Answer:- (Page 50)

A bridge is a hardware device also used to connect two LAN segments to extend a LAN. A typical bridge has two NICs, a CPU a memory and a ROM. It only runs the code stored in its ROM.

**ambiguity may cause the failure of any network, how parity bit check is consider in this situation
5 marks**

define default route how it is different from routing 5 marks

CS610- Computer Network Midterm Examination - Spring 2011

Question: Defines simplex and full duplex connection? Marks 2

Answer:- (Page 76)

Some connection-oriented technologies provide full duplex while other allow on simplex connection. To communicate using a simplex design a pair of computers must establish two connections one from computer A to computer B and another from computer B to A.

Question: How WAN capable to handle many computers? Marks 2

Answer:- [Click here for detail](#)

A Wide Area Network (WAN) is a telecommunication network that covers a broad area.

Question: Briefly explains the working structure of thick Ethernet. Marks 2

Answer:- (Page 41)

It uses thick coax cable. AUI cable (or transceiver or drop cable) connects from NIC to transceiver. AUI cable carries digital signal from NIC to transceiver. The transceiver generates analog signal on coax cable. The wires in AUI carry digital signals power and other control signals. Thick Ethernet also requires terminators to avoid signal reflectance.

Question: Imagines a network for six devices: A, B, C, D, E and F. all devices are on a same packet switch. Computer A want to send packet to computer E. how would packet switch forward this packet? Marks 3

Answer:- rep

Question: Does ATM eliminate the varying delay times associated with different sized packets? If yes then give reason. Marks 2

Question: Why ATM design chose cells over packet? Marks 5

Answer:- (Page 72)

- Cells are not variable length and memory management for them is simpler. Handling variable length packets leads to memory fragmentation.
- Variable length packets require hardware to accommodate the largest possible packet, and thus to detect the end of the packet. With cells bits can just be counted as they arrive.
- The length of time required to send a variable length packet is variable and requires complicated interrupt scheme to detect completion of transmission. QoS can't be guaranteed with variable length packets as easily as it can with fixed length cells.

Question: Reception is a process of CPU; describe it in a few lines? Marks 5

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What is 10base T?

Answer:- [Click here for detail](#)

One of several adaptations of the Ethernet (IEEE 802.3) standard for Local Area Networks (LANs). The 10Base-T standard (also called Twisted Pair Ethernet) uses a twisted-pair cable with maximum lengths of 100 meters.

What is Multicasting?

Answer:- (Page 35)

It works like broadcasting however it does not forward frames automatically to the CPU.

The interface hardware is programmed in advance to accept certain frames that have multicast address as the destination address.

What is FDDI?

Answer:- (Page 31)

Fiber distributed data interconnect (FDDI) is another ring technology. Its most important features are:

- ❖ It uses fiber optics between stations and transmits data at 100Mbps.
- ❖ It uses pair of fibers to form two concentric rings.

Define 802.11 Wireless lans and CSMA?

Answer:- (Page 29)

IEEE 802.11 is standard wireless LAN that uses radio signals at 2.4GHz. Its data rate is 11Mbps. The older devices use radio signals at 900MHz and data rate of 2Mbps. Bluetooth specifies a wireless LAN for short distances. It uses shared medium and radio waves instead of coaxial cable.

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Define static and dynamic routing.(2)

Answer:- [Click here for detail](#)

Static Routing:-

Static routing is the simplest form of routing, but it is a manual process.

Dynamic Routing:-

Dynamic routing protocols are supported by software applications running on the routing device (the router) which dynamically learn network destinations and how to get to them and also advertise those destinations to other routers.

Dijkart's Algorithm 2 Marks

Answer:- rep

write note on Bridges 5 Marks

Answer:- rep

Disadvantages of Repeaters 3 Marks

Answer:- (Page 50)

Repeaters do not recognize frame formats, they just amplify and retransmit the electrical signal. If a collision or error occurs in one segment, repeaters amplify and retransmit also the error onto the other segments.

Different b/w Cells and Packets 5 Marks

Answer:- (Page 72)

ATM designers chose cells over packets because of the following reasons:

- Cells are not variable length and memory management for them is simpler. Handling variable length packets leads to memory fragmentation.
- Variable length packets require hardware to accommodate the largest possible packet, and thus to detect the end of the packet. With cells bits can just be counted as they arrive.
- The length of time required to send a variable length packet is variable and requires complicated interrupt scheme to detect completion of transmission. QoS can't be guaranteed with variable length packets as easily as it can with fixed length cells.

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21.What is the difference between the physical and logical topologies?(2)

Answer:-

Physical Topology:

The way that the workstations are connected to the network through the actual cables that transmit data -- the physical structure of the network -- is called the physical topology. It depends on the wiring scheme.

Logical Topology:

The logical topology, in contrast, is the way that the signals act on the network media, or the way that the data passes through the network from one device to the next without regard to the physical interconnection of the devices. We can say that it is defined by the specific network technology.

22. Define static and dynamic routing.(2)

Answer:- rep

23. What is meant by Bridges STARTUP and STEADY State?(3)

Answer:- (Page 51)

When a bridge first boots the address lists are empty (start up state). The bridge forwards frames to the other segment if it can not find its destination address in its lists.

After some time when the bridge has received at least one frame from every computer, it has the lists built (steady state) it forwards frames as far it is necessary.

24. How can Switched Virtual Network be established?(3)

Answer:- (Page 70)

ESTABLISHING AN SVC:

The computer sends a connection request to the switch to which it is attached. Software in the switch finds a network path to the destination and sends along the connection request.

Each pair of switches in the path communicates to choose a VPI/VCI for their tables. Once the connection is established by the destination, a message is sent back to the originating computer to indicate the SVC is ready.

If any switch or the destination computer does not agree to setting up the VC, an error message is sent back and the SVC is not established

25. What is the concept of packet switches?(5)

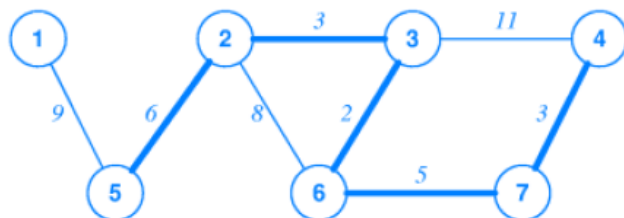
Answer:- (Page 55)

To span long distances or many computers, networks must replace shared medium with packet switches. Each switch moves an entire packet from one connection to another. That's why they are called packet switches. A packet switch consists of a small computer with network interfaces, a memory and a program dedicated to packet switching function.

26. Write a note on Weight Graph.(5)

Answer:- (Page 62)

Dijkstra's algorithm can accommodate weights on edges in graph. The shortest path is then the path with lowest total weight (sum of the weight with all edges). It should be noted that the shortest path is not necessarily with fewest edges (or hops). For example as shown in the figure below:



The shortest path in the figure from node 2 to node 6 is 2 to 3 and 3 to 6 as this path has the smallest weight so it is the shortest path.

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Question No: 21 (M a r k s: 2)

What is the difference between the physical and logical topologies?

Answer:- rep

Question No: 22 (M a r k s: 2)

Define Vector-Distance Algorithm.

Answer:- (Page 155)

Packet switches wait for next update message and they iterate through entries in message. If entry has shortest path to destination, insert source as next hop to destination and record distance as distance from next hop to destination plus distance from this switch to next hop.

Question No: 23 (M a r k s: 3)

What is the concept of store and forward technology?

Answer:- (Page 56)

STORE AND FORWARD:

Data delivery from one computer to another is accomplished through store and forward technology. In this technology packet switch stores incoming packet and also forwards that packet to another switch or computer. For this purpose packet switch has internal memory into which it can hold packet if outgoing connection is busy. Packets for each connection held on queue

Question No: 24 (M a r k s: 3)

How can Switched Virtual Network be established?

Answer:- rep

Question No: 25 (M a r k s: 5)

How can a bridge know whether to forward frames?

Answer:- (Page 53)

The bridges configure themselves automatically to decide which bridge will forward broadcast frames and which bridge will not.

The bridges communicate with each other on the network and use Distributed Spanning Tree (DST) algorithm to decide which bridge will not forward frames if a cycle occurs.

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Question No: 26 (Marks: 5)

Compare connection oriented and connectionless Service.

Answer:- [Click here for detail](#)

- ❖ In connection-oriented protocol, authentication is needed while this is not case in connectionless protocol.
- ❖ In connection-oriented protocol, we have to establish connection between sender and receiver while this is not case in connectionless protocol.
- ❖ Example of connection-oriented protocol is TCP and the example of connectionless protocol is UDP, Internet.
- ❖ TCP is a connection-oriented protocol, it makes a connection and checks whether the data is received, and resends if it is not. UDP is a connectionless protocol, it does not guarantee delivery by first connecting and checking whether data is received.

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Question No: 21 (Marks: 2)

What is the difference between LAN and WAN?

Answer:- rep

Question No: 22 (Marks: 2)

Define the term Jitter.

Answer:- rep

Question No: 23 (Marks: 3)

Give a comparison of wiring Schemes.

Answer:- (Page 44)

The wiring schemes are compared as follows:

Separate transceiver allows computers to be powered off or disconnected from network without disrupting other communication.

Transceiver may be located in an inconvenient place, so finely malfunction transceiver can be hard.

In other case, thin coax cable takes minimum of cable. Disconnecting one computer (on one loose connection) can disrupt entire network.

Hub wiring centralizes electronics and connections. It makes management easier. Bottom line 10Base-T is most popular because of lowest cost.

Question No: 24 (Marks: 3)

How can Switched Virtual Network be established?

Answer:- rep

Question No: 25 (Marks: 5)

Describe permanent virtual circuits (PVC).

Answer:- (Page 69)

ATM can provide customers with virtual circuits that look like traditional leased digital circuits. Such permanent virtual circuits (PVC) last as long as the customer pay the periodic fee for its use. The forwarding tables are automatically restored after power of equipment failure. The forwarding table entries for such permanent VC's are statically configured, the terms used by Telco's for this is provisioning.

Provisioning requires two steps:

1. To determine a complete path (that is, identify the switches that will be used).
2. To choose appropriate VPI/VCI for each step in the path, and configure each adjacent pair of switches (easy, since each switch rewrites the VCI/VPI).

Question No: 26 (Marks: 5)

What are default routes draw the table.

Answer:- (Page 60)

Routing table entries can be collapsed with a default route. If the destination does not have in explicit routing table entry, then it uses a default route. Default routes for 4 nodes are shown in the figure below.

destin- ation	next hop	destin- ation	next hop	destin- ation	next hop	destin- ation	next hop
1	-	2	-	1	(3,1)	2	(4,2)
*	(1,3)	4	(2,4)	2	(3,2)	4	-
		*	(2,3)	3	-	*	(4,3)
				4	(3,4)		
<i>node 1</i>		<i>node 2</i>		<i>node 3</i>		<i>node 4</i>	



CS610- Computer Network

Solved MCQS
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Question No: 1 (M a r k s : 1)

No error detection scheme is perfect because transmission errors can affect the additional information as well as the data.

► True (Computer Networks and Internets, page82)

► False

Question No: 2 (M a r k s : 1)

----- Program sends a message to a remote computer and reports whether the computer responds.

► Ping (Computer Networks and Internets, page9)

► Traceroute

► ICMP

► Non of the given

Question No: 3 (M a r k s : 1)

----- was especially concerned about the lack of high powered computers.

► ARPA (Computer Networks and Internets, page5)

► IEEE

► EIA

► Non of the given

Question No: 4 (M a r k s : 1)

The term ----- is used to denote the definition of a packet used with a specific type of network.

► Packet

► Frame (Computer Networks and Internets, page 73)

► Data

► None of the given

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Question No: 5 (M a r k s: 1)

Computer networks are often called ----- because they use packet technology.

- ▶ Ethernet
- ▶ Switch networks
- ▶ [Packet networks](#) (Computer Networks and Internets, page 73)
- ▶ None of the given

Question No: 6 (M a r k s: 1)

----- have advantages arisen from the size and ease of computation.

- ▶ CRC
- ▶ [Parity](#) (Page 19)
- ▶ Checksums
- ▶ None of given

Question No: 7 (M a r k s: 1)

Most LANs that employ ring topology use an access mechanism known as-----

- ▶ CSMA/CD
- ▶ CSMA/CA
- ▶ [TOKEN PASSING](#) (Computer Networks and Internets, page95)
- ▶ None of the given

Question No: 8 (M a r k s: 1)

IEEE LLC/SNAP header is -----, which is used to specify the type of data.

- ▶ [8 octets](#) (Computer Networks and Internets, page112)
- ▶ 8 bytes
- ▶ 8 bits
- ▶ None of the given

Question No: 9 (M a r k s: 1)

Formally named _____ informally known as the twisted pair Ethernet or TP Ethernet.

- ▶ 10 Base 2
- ▶ 10 Base 5
- ▶ [10 Base T](#) (Page 43)
- ▶ None of the given

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Question No: 10 (M a r k s: 1)

An interface for twisted pair Ethernet must have an _____ connector , and must generate signals according to the _____ specification.

▶ [RJ-45, 10 Base T](#) (Computer Networks and Internets, page 126)

- ▶ RJ-45, 10 Base 5
- ▶ BNC, 10 Base 2
- ▶ BNC, 10 Base T

Question No: 11 (M a r k s: 1)

A bridges function in the _____ layers(s).

▶ [Physical \(MAC\)](#) [Click here for Detail](#)

- ▶ Data link
- ▶ Network
- ▶ Physical (MAC) and Data link

Question No: 12 (M a r k s: 1)

A Bridge can _____

- ▶ Filter a frame
- ▶ Forward a frame
- ▶ Extend a LAN

▶ [Do all the above](#) [Click here for Detail](#)

Question No: 13 (M a r k s: 1)

A Bridge forwards or filters a frame by comparing the information in its address table to the frame's _____

- ▶ Layer 2 source address
- ▶ Source node's physical address

▶ [Layer 2 destination address](#) [Click here for Detail](#)

- ▶ Layer 3 destination address

Question No: 14 (M a r k s: 1)

_____ computes shortest paths in a graph by using weights on edges as a measure of distance.

- ▶ Greedy algorithm
- ▶ Distance vector algorithm

▶ [Dijkstra's algorithm](#) (Computer Networks and Internets, page112)

- ▶ Non of the given

Question No: 15 (M a r k s: 1)

_____ is used for audio and video, since these have predefined maximum data rates

► **Constant Bit Rate (CBR) service** (Page 71)

- Variable Bit Rate (VBR) service
- Available Bit Rate (ABR) service
- None of the given

Question No: 16 (M a r k s: 1)

Unlike Frame Relay and ATM, SMDS (Switched multi-megabit Data service) offers_____ .

► **Connectionless service paradigm** (Computer Networks and Internets, page112)

- Connection oriented service paradigm
- Both Connectionless and Connection-oriented service paradigm
- None of the given

Question No: 17 (M a r k s: 1)

A network with throughput T and delay D has a total of _____ bit in transit at any time.

- T / D

► **T x D** (Computer Networks and Internets, page203)

- T + D
- None of the given

Question No: 18 (M a r k s: 1)

ATM is designed to work on_____.

- Twisted Pair
- Coaxial
- Radio Frequency

► **Fiber** (Computer Networks and Internets, page 72)

Question No: 19 (M a r k s: 1)

Computers attached to an Ethernet use ----- in which a computer waits for the ether to be idle before transmitting a frame.

► **CSMA/CD** (Computer Networks and Internets, page 93)

- CSMA/CA
- TOKEN PASSING
- None of the given

Question No: 20 (M a r k s: 1)

FDDI can transmits data at a rate of -----

▶ **100 million bits per second** (Page 314)

- ▶ 10 million bits per second
- ▶ 1000 million bits per second
- ▶ None of the given

**CS610- Computer Network
MIDTERM EXAMINATION
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Question No: 1 (M a r k s: 1)

In Point-to-Point topology there are two topologies.

- ▶ Star and Tree
- ▶ Tree and Ring
- ▶ Star and Ring

▶ **Star and Tree** (Page 5)

- ▶ None of the given

Question No: 2 (M a r k s: 1)

----- Program sends a message to a remote computer and reports whether the computer responds.

▶ **Ping** (Computer Networks and Internets, page9)

- ▶ Traceroute
- ▶ ICMP
- ▶ Non of the given

Question No: 3 (M a r k s: 1)

----- has no way to determine the cause of the problem.

- ▶ ICMP

▶ **Ping** (Computer Networks and Internets, page 11)

- ▶ Trace route
- ▶ Non of the given

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Question No: 4 (M a r k s: 1)

The term----- refers to the general concept of a small block of data

► **Packet** (Page 16)

- Frame
- Data
- None of the given

Question No: 5 (M a r k s: 1)

----- scheme, which is designed to help detect transmissions errors, send one extra bit of information with each character

► **Parity** (Computer Networks and Internets, page 77)

- Checksums
- CRC
- None of given

Question No: 6 (M a r k s: 1)

Local Talk is a LAN technology that employs -----

► **Bus topology** (Page 30)

- Ring topology
- Star topology
- None of the given

Question No: 7 (M a r k s: 1)

Most LANs that employ ring topology use an access mechanism known as-----

- CSMA/CD
- CSMA/CA

► **TOKEN PASSING** (Computer Networks and Internets, page95) rep

- None of the given

Question No: 8 (M a r k s: 1)

Ethernet uses a ----- bit static addressing scheme in which each device is assigned a unique address by the manufacturer.

- 64

► **48** (Computer Networks and Internets, page 109)

- 32
- 8

Question No: 9 (M a r k s: 1)

Formally named _____ informally known as the thick wire Ethernet or Thick net.

- ▶ 10 Base 2
- ▶ [10 Base 5](#) **Computer Networks and Internets, page 120)**
- ▶ 10 Base T
- ▶ None of the given

Question No: 10 (M a r k s: 1)

Formally named _____ informally known as the twisted pair Ethernet or TP Ethernet.

- ▶ 10 Base 2
- ▶ 10 Base 5
- ▶ [10 Base T](#) **(Page 43) rep**
- ▶ None of the given

Question No: 11 (M a r k s: 1)

The maximum size of an Ethernet segment is _____

- ▶ 250 meters
- ▶ [500 meters](#) **(Page 27)**
- ▶ 700 meters
- ▶ None of the given

Question No: 12 (M a r k s: 1)

A Bridge can _____

- ▶ Filter a frame
- ▶ Forward a frame
- ▶ Extend a LAN
- ▶ [Do all the above](#) **rep** [Click here For Detail](#)

Question No: 13 (M a r k s: 1)

_____ computes shortest paths in a graph by using weights on edges as a measure of distance.

- ▶ Greedy algorithm
- ▶ Distance vector algorithm
- ▶ [Dijkstra's algorithm](#) **(Computer Networks and Internets, page112) rep**
- ▶ Non of the given

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Question No: 14 (M a r k s: 1)

_____ is used for compressed audio and video where the data rate depends on the level of compression that can be achieved.

- ▶ Constant Bit Rate (CBR) service
- ▶ [Variable Bit Rate \(VBR\) service](#) (Page 71)
- ▶ Available Bit Rate (ABR) service
- ▶ None of the given

Question No: 15 (M a r k s: 1)

Basic LAN technologies such as Ethernet, Token Ring, and FDDI use a _____.

- ▶ [Connectionless service paradigm](#) (Computer Networks and Internets, page112)
- ▶ Connection-oriented service paradigm
- ▶ Both Connectionless and Connection-oriented service paradigm
- ▶ None of the given

Question No: 16 (M a r k s: 1)

The product of delay and throughput measures the _____ of data that can be present on the network.

- ▶ Area
- ▶ [Volume](#) (Page 80)
- ▶ Length
- ▶ None of the given

Question No: 17 (M a r k s: 1)

A network with throughput T and delay D has a total of _____ bit in transit at any time.

- ▶ T / D
- ▶ [T x D](#) (Computer Networks and Internets, page203) rep
- ▶ $T + D$
- ▶ None of the given

Question No: 18 (M a r k s: 1)

One repeater _____, two repeaters _____ the maximum cable length limitation.

- ▶ [doubles, triple](#) (Page 49)
- ▶ square roots, cube roots
- ▶ and, triple
- ▶ doubles, cancel

Question No: 19 (Marks: 1)

A network uses a ----- if all computers attach to a central point

► **Star Topology** (Page 25)

- Ring Topology
- Bus Topology
- None of the given

Question No: 20 (Marks: 1)

Computers attached to an Ethernet use ----- in which a computer waits for the other to be idle before transmitting a frame.

► **CSMA/CD** (Computer Networks and Internets, page 93) rep

- CSMA/CA
- TOKEN PASSING
- None of the given

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Question No: 1 (Marks: 1) - Please choose one

A typical port on an ATM switch operates at _____ or higher.

- OC-2 speed (155Mbps)
- OC-3 speed (100Mbps)

► **OC-3 speed (155Mbps)** (Page 72)

- OC-3 speed (155Gbps)

Question No: 2 (Marks: 1) - Please choose one

The product of delay and throughput measures the _____ of data that can be present on the network.

- Area

► **Volume** (Page 80) rep

- Length
- None of the given

Question No: 3 (Marks: 1) - Please choose one

_____ is used for compressed audio and video where the data rate depends on the level of compression that can be achieved.

- Constant Bit Rate (CBR) service

► **Variable Bit Rate (VBR) service** (Page 71) rep

- Available Bit Rate (ABR) service
- None of the given

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Question No: 4 (Marks: 1) - Please choose one

Which of the following is a connecting device?

- ▶ Bridge
- ▶ Repeater
- ▶ Hub

▶ [All the given](#) [Click here for Detail](#)

Question No: 5 (Marks: 1) - Please choose one

A bridges function in the _____ layers(s).

▶ [Physical \(MAC\)](#) rep [Click here for Detail](#)

- ▶ Data link
- ▶ Network
- ▶ Physical (MAC) and Data link

Question No: 6 (Marks: 1) - Please choose one

IEEE LLC/SNAP header is -----, which is used to specify the type of data.

▶ [8 octets](#) (Computer Networks and Internets, page112) rep

- ▶ 8 bytes
- ▶ 8 bits
- ▶ None of the given

Question No: 7 (Marks: 1) - Please choose one

The third field of the header consists of ----- bit Ethernet frame type.

- ▶ 48
- ▶ 32

▶ [16](#) (Computer Networks and Internets, page110)

- ▶ 8

Question No: 8 (Marks: 1) - Please choose one

An -----method, the network hardware designers specify how type information is included in the frame and the value use to identify various frame types.

▶ [Explicit frame type](#) (Page 35)

- ▶ Ideal frame type
- ▶ Implicit frame type
- ▶ None of the given

Question No: 9 (Marks: 1) - Please choose one

Local Talk is a LAN technology that employs -----

- ▶ Star topology

▶ [Bus topology](#) (Page 30) rep

- ▶ Ring topology
- ▶ None of the given

Question No: 10 (Marks: 1) - Please choose one

The Fast Ethernet hardware operates at a rate of -----

- ▶ 10 Mbps
- ▶ [100 Mbps](#) (Page 27)
- ▶ 1000 Mbps
- ▶ None of the given

Question No: 11 (Marks: 1) - Please choose one

----- scheme, which is designed to help detect transmissions errors, send one extra bit of information with each character

- ▶ [Parity](#) (Computer Networks and Internets, page 77) rep
- ▶ Checksums
- ▶ CRC
- ▶ None of given

Question No: 12 (Marks: 1) - Please choose one

Computer networks are often called ----- because they use packet technology.

- ▶ Ethernet
- ▶ Switch networks
- ▶ [Packet networks](#) (Computer Networks and Internets, page 73) rep
- ▶ None of the given

Question No: 13 (Marks: 1) - Please choose one

----- Program sends a message to a remote computer and reports whether the computer responds.

- ▶ [Ping](#) (Computer Networks and Internets, page 9) rep
- ▶ Traceroute
- ▶ ICMP
- ▶ Non of the given

Question No: 14 (Marks: 1) - Please choose one

In -----, network occupies larger areas like cities & countries.

- ▶ LAN
- ▶ [WAN](#) (Page 4)
- ▶ MAN
- ▶ None of the given

Question No: 15 (Marks: 1) - Please choose one

In -----, network occupies the smaller area like a room a floor or a building

- ▶ [LAN](#) (Page 4)
- ▶ WAN
- ▶ MAN
- ▶ None of the given

Question No: 16 (Marks: 1) - Please choose one

In Point-to-Point topology there are two topologies.

- ▶ Tree and Ring
- ▶ Star and Ring
- ▶ Star and Tree (Page 5)
- ▶ None of the given

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Question No: 1 (Marks: 1) - Please choose one

_____ has a jitter zero

- ▶ None of the given
- ▶ Virtual Private Network
- ▶ Isochronous Network
- ▶ Asynchronous Network (Page 66)

Question No: 2 (Marks: 1) - Please choose one

Unlike Frame Relay and ATM, SMDS (Switched multi-megabit Data service) offers_____ .

- ▶ Connectionless service paradigm (Computer Networks and Internets, page112)
- ▶ Connection oriented service paradigm
- ▶ Both Connectionless and Connection-oriented service paradigm
- ▶ None of the given

Question No: 3 (Marks: 1) - Please choose one

ATM assigns each VC a _____ identifier that is divided two parts to produce a hierarchy.

- ▶ 21-bit
- ▶ 22-bit
- ▶ 23-bit
- ▶ 24-bit (Page 67)

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Question No: 4 (Marks: 1) - Please choose one

Most WAN systems include a mechanism that can be used to eliminate the common case of duplication routing is called_____

- ▶ Hierarchal address
- ▶ [Default route](#) (Computer Networks and Internets, page172)
- ▶ Shortest path
- ▶ None of the given

Question No: 5 (Marks: 1) - Please choose one

The next hop to which a packet is sent depends only on

- ▶ [Packet's destination](#) (Computer Networks and Internets, page169)
- ▶ Packet's original source
- ▶ Path the packet has taken
- ▶ Non of the given

Question No: 6 (Marks: 1) - Please choose one

An interface for twisted pair Ethernet must have an _____ connector , and must generate signals according to the_____ specification.

- ▶ [RJ-45, 10 Base T](#) (Computer Networks and Internets, page 126) rep
- ▶ RJ-45, 10 Base 5
- ▶ BNC, 10 Base 2
- ▶ BNC, 10 Base T

Question No: 7 (Marks: 1) - Please choose one

When an application----- data, it makes a copy of the data available to all other computers on the network.

- ▶ [Broadcasting](#) [Click here for Detail](#)
- ▶ Multicasting
- ▶ Unicasting
- ▶ None of the given

Question No: 8 (Marks: 1) - Please choose one

A ----- provide a mechanism that a customer can use to set a physical address.

- ▶ Static addressing scheme
- ▶ [Configurable addressing scheme](#) (Page 34)
- ▶ Dynamic addressing scheme
- ▶ None of the given

Question No: 9 (Marks: 1) - Please choose one

FDDI can transmits data at a rate of -----

▶ [100 million bits per second](#) (Page 314) rep

- ▶ 100 million bits per second
- ▶ 100 million bits per second
- ▶ None of the given

Question No: 10 (Marks: 1) - Please choose one

Computers attached to an ether use ----- in which a computer waits for the ether to be idle before transmitting a frame.

▶ [CSMA/CD](#) (Computer Networks and Internets, page 93) rep

- ▶ CSMA/CA
- ▶ TOKEN PASSING
- ▶ None of the given

Question No: 11 (Marks: 1) - Please choose one

----- have advantages arisen from the size and ease of computation.

▶ CRC

▶ [Parity](#) (Page 19) rep

- ▶ Checksums
- ▶ None of given

Question No: 12 (Marks: 1) - Please choose one

The term ----- is used to denote the definition of a packet used with a specific type of network.

▶ Packet

▶ [Frame](#) (Computer Networks and Internets, page 73)

- ▶ Data
- ▶ None of the given

Question No: 13 (Marks: 1) - Please choose one

----- has no way to determine the cause of the problem.

▶ [Ping](#) (Computer Networks and Internets, page 11) rep

- ▶ Trace route
- ▶ ICMP
- ▶ Non of the given

Question No: 14 (Marks: 1) - Please choose one

----- Program sends a message to a remote computer and reports whether the computer responds.

▶ Ping (Computer Networks and Internets, page9) rep

- ▶ Traceroute
- ▶ ICMP
- ▶ Non of the given

Question No: 15 (Marks: 1) - Please choose one

In -----, network occupies the smaller area like a room a floor or a building

▶ LAN (Page 4) rep

- ▶ WAN
- ▶ MAN
- ▶ None of the given

Question No: 16 (Marks: 1) - Please choose one

No error detection scheme is perfect because transmission errors can affect the additional information as well as the data.

▶ True (Computer Networks and Internets, page82)

- ▶ False

Helping Material

Cs610- Computer Networks By Hammad Khalid Khan

Computer Networks and Internets with Internet Applications, 4/e (With CD) By Comer

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